PV investigation
Paired development evidence

19 OCTOBER 2026 / CONTROLLED FIRST SLICE

Equal physical outcomes; the investigation mechanism is now observable.

The procedural reference (R) and bounded goal planner (A) completed four matched development cases: **8/8 episodes met the frozen success rubric**. The records expose sensing, external questions, decisions, predictions, physical execution and fresh verification on the same 16-second simulation horizon.

Measured result	This batch
Cases and repetitions	4 cases x 2 arms x 1 episode; one declared seed
Prediction outcomes	16/16 numerical precommitments verified
Recorded rubric errors	0
Exact record replay	8/8 metric-equal replays
Live information calls / new camera frames	0 / 0
Matched PV energy	Equal in all four pairs

C01: unchanged asset, known hold instruction. **C02:** hidden light reduction, known hold instruction. **C03:** missing service context, returned hold instruction. **C04:** the same local physical evidence, returned permission to survey.

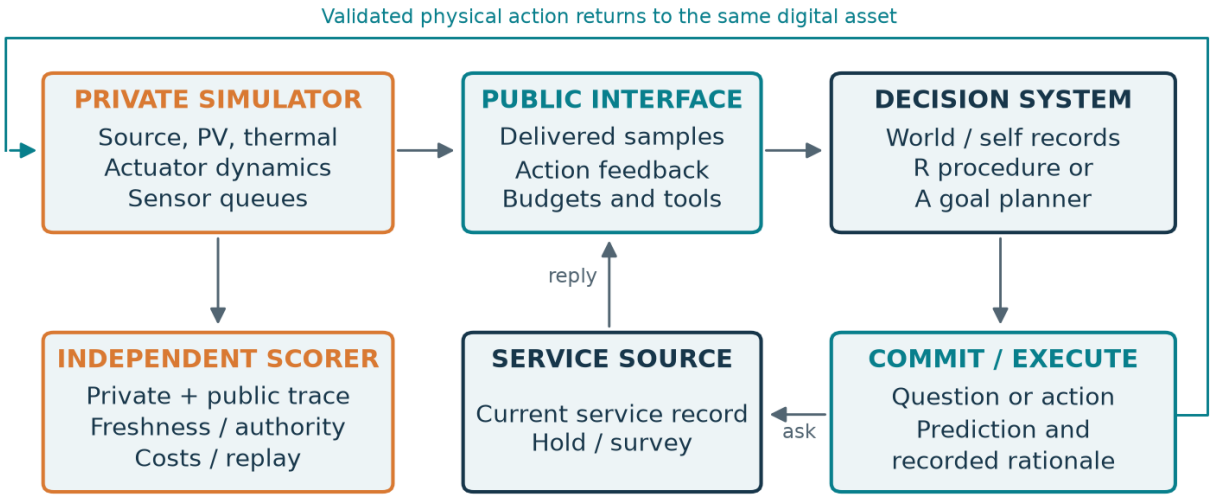
What the outcome means

This is evidence that the common investigation loop can execute, retain its reasoning state and verify a permitted action. The competent reference succeeds too. These cases therefore establish an auditable comparison and **do not establish an adaptive performance advantage**, open-ended intelligence or transfer to a real PV asset.

How to read the evidence

Orange denotes evaluator-only knowledge; teal denotes public evidence or action. Every 3D view in this report is a reconstruction from recorded geometry, not a newly captured camera image. The paired interactive viewer is the companion for inspecting the same decisions at their actual availability times.

1. The two systems and the evidence boundary



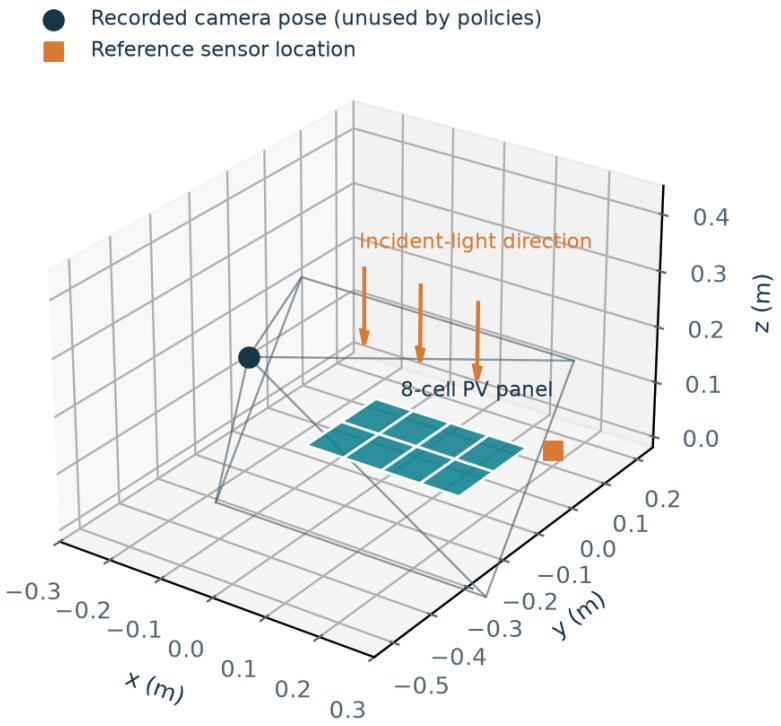
Recorded evidence, not private state, is the input to both policies.

Component	Implemented role
Environment world model	Private source, optics, piecewise electrical model, thermal dynamics, actuators and measurement timing generate the digital world.
Agent world belief	A source-linked summary of reference evidence, unresolved physical cause, current service permission and measured operating target.
Agent self model	Advertised tools, remaining requests / probe points, observed failures and the requirement for fresh V/I verification.
R: procedural reference	Fixed observe -> resolve missing permission -> survey or hold -> verify procedure, with competent measured branches and bounded clarification.
A: bounded goal planner	Re-evaluates unsatisfied goals and eligible operators after each returned context; fixed explicit priorities choose the next operation.

Both arms use the same evidence-validation helpers, public brief, tools, source record, seed and physical configuration. Neither receives the private case label or hidden source schedule. A is deterministic symbolic planning: there is no language model, learned policy, probabilistic belief calibration, cross-episode memory or image interpretation.

2. The digital asset, sensors and 3D scene

Evaluator reconstruction from C04-A config.json and recorded truth; geometry is not policy input.



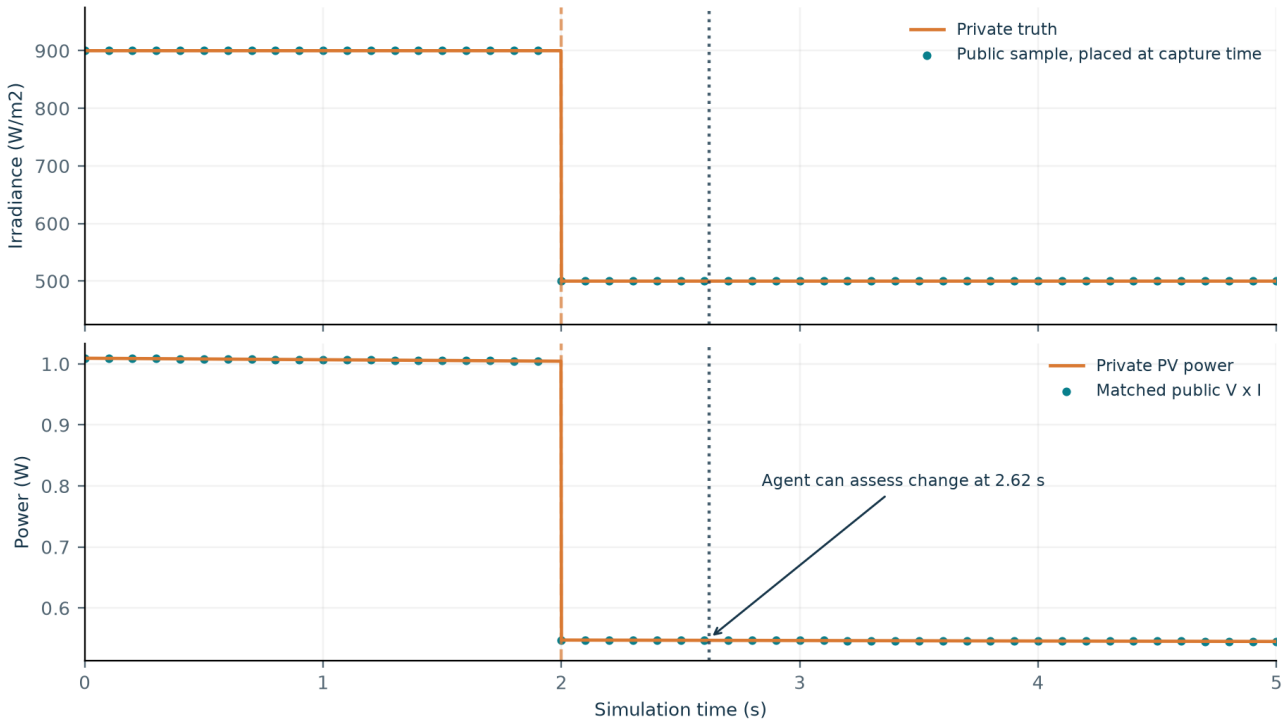
The frozen scene contains a 0.30 x 0.20 m panel with eight cells arranged in two series substrings, a reference irradiance sensor and the recorded camera mechanism. The frustum uses the camera's recorded position and rotation plus its declared field of view. Rays illustrate the configured source direction; they do not imply a measured light-source position.

Digital mechanism	What this comparison exercises
Incident light and PV response	A source step changes irradiance and the piecewise-cell current/power response; thermal state evolves with input and cooling.
Voltage actuator	Delay, slew and settling separate a requested voltage from its measured effect. A three-point probe returns settled V/I confirmations.
Public numeric sensing	10 Hz scalar captures; 0.02 s numeric delivery latency. The policy receives batches only when an action returns.
Camera / RGB	The scene has a camera pose, but the batch runs with rendering disabled and records zero new RGB frames. No camera decision or visual diagnosis is tested.

The pilot disables numeric noise and actuator failures. This controlled simulator evidence tests internal digital consistency, not agreement with calibrated hardware.

3. A hidden physical change becomes evidence

C02-A: source schedule is evaluator-only; the policy learns from delivered measurements.

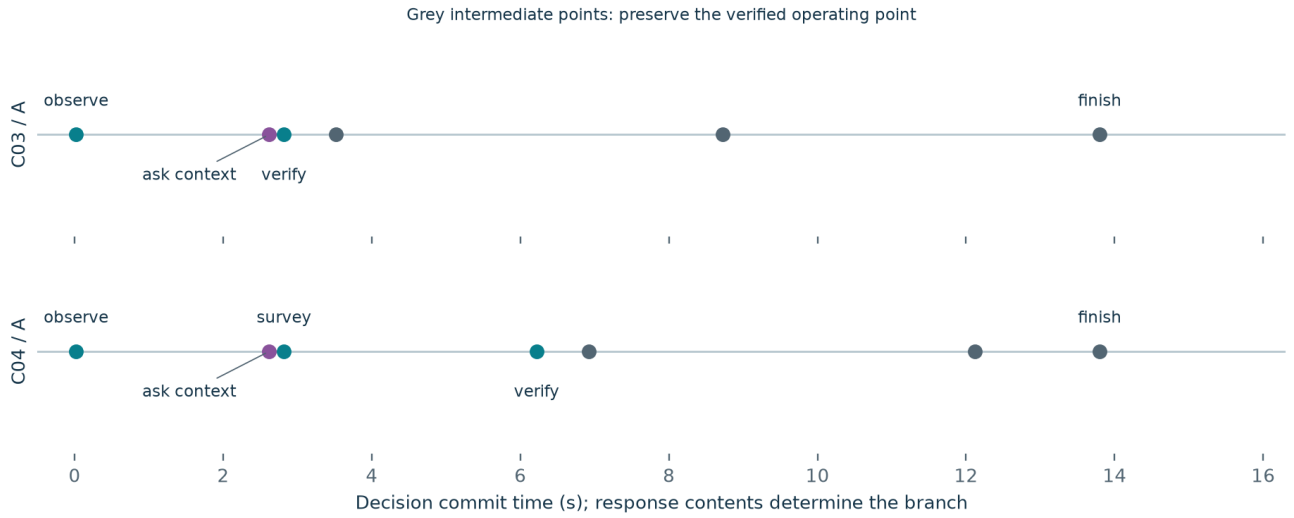


At 2.00 s the hidden direct component changes from 800 to 400 W/m2 while diffuse light remains 100 W/m2. The reference therefore changes from 900 to 500 W/m2. The first changed public capture is at **2.00 s**; the policy can use it at **2.62 s**, when the local observation action returns. The chart places public points at capture time only to compare their values with truth; they were not visible to the agent at all those earlier times.

Step	Recorded interpretation and decision
Sense	Compare valid reference samples: 900.0 to 500.0 W/m2; threshold is 100 W/m2.
Analyse	Report measured incident-light reduction. Do not turn this into an unsupported claim of dirt, panel damage, cloud or a unique external cause.
Act	The public instruction already says hold 1.8 V. Both arms avoid an unnecessary external query or voltage survey.
Verify	Require matched valid voltage/current captured after the relevant action completed; preserve the unique physical cause as unresolved.

4. Why external information changes the action

C03 and C04 share the same local physical evidence before the query. Only the service record changes.



Actual A question

Local reference evidence is no_threshold_reduction; this does not establish a physical cause. What current service instruction governs whether I should hold 1.8 V or perform the declared bounded voltage survey?

Returned authoritative record	Decision consequence
C03: hold	Preserve nominal voltage; collect fresh matched V/I after the reply; no survey.
C04: survey	Run the permitted 0.8, 1.8 and 3.0 V survey; rank measured settled power; apply the best candidate if necessary; verify.
Counterfactual check	Pre-query local-history SHA-256 is identical between C03 and C04 for each arm. The branch cannot be explained by different local measurements.

What this source is

The internet-labelled channel calls a **controlled current-record lookup**. It accepts newly worded questions but returns the same corpus record for that case; it performs no semantic web search or human reasoning. Typed current hold/survey fields and declared fixture source identity determine permission. The original text is retained. Source-title matching is not cryptographic authentication, and prose/typed-field disagreement is not interpreted semantically.

External waiting is recorded in wall-clock time while the simulator is paused. This protocol prevents fictitious sunlight changes during a local fixture lookup; it does not measure real-world energy loss from a slow human or internet reply.

5. The agent's recorded model and choice

C04-A at the decision after the controlled survey permission is received.

World belief at 2.82 s

Reference evidence: no_threshold_reduction; unique physical cause unresolved. Current service instruction: survey (current_authorized_service_record). Operating target from public evidence: 1.800 V.

Self model

Advertised tools remain bounded by public limits. 2 external requests remain; 12 voltage points remain. Failed physical actions observed: 0. Completion needs fresh matched V/I, not acknowledgement alone.

Alternatives actually evaluated

Operator	Eligible / priority	Recorded reason
safe_stop	no / 1000	Failed or unverified execution cannot support a successful conclusion.
observe	no / 100	The local evidence window is incomplete; another source cannot replace it.
ask_context	no / 90	Local sensors cannot fill the decision-relevant service-permission gap.
survey	yes / 80	Permission allows experiments; measured candidate power ranking is missing.
apply_target	no / 70	The greatest measured-power candidate is not the latest measured operating point.
verify	yes / 60	Fresh matched V/I after the last change must test the expected effect.
safe_stop	no / 50	Bounded additional observation did not establish successful application.

The selected action is determined by the highest declared priority among eligible operators. Priorities are design choices, not learned value-of-information estimates. The record makes the next-step selection inspectable; it is not a private chain of thought or an after-the-fact explanation.

What differs from the reference

R reaches a competent measured branch through its fixed procedure. A recomputes goals and eligibility at each context boundary. On these simple cases the resulting choices are equivalent. Demonstrating replanning under interruption or failure requires additional cases that challenge the procedure without handicapping its evidence checks.

6. Predict, execute, then verify

C04-A: the voltage survey creates measured evidence; a later commitment tests the applied target.

Survey target (V)	Measured V	Measured I (A)	Measured power (W)	Captured / usable (s)
0.8	0.800	0.74838	0.59870	3.80 / 6.22
1.8	1.800	0.55438	0.99788	4.70 / 6.22
3.0	3.000	0.31918	0.95755	5.70 / 6.22

The selected candidate is **1.8 V**, based on the greatest settled measured V x I. It is the best of three measured candidates, not proof of the continuous global maximum-power point. The policy does not manufacture a current prediction before the probe: that prediction is explicitly marked not predictable.

Here the probe restores the nominal 1.8 V operating point, which is also the measured best candidate. Both arms avoid a redundant set-voltage command and perform a fresh verification wait. A separate regression forces a different best candidate to test application and verification; this batch does not show a benefit from adopting a new voltage.

Prediction	Commit (s)	Target capture (s)	Observed / tolerance	Usable (s)
decision_001_voltage	0.02	2.80	1.800 V / +/- 0.02 V	2.82
decision_004_voltage	6.22	7.10	1.800 V / +/- 0.02 V	12.12

Two checks occur at different times. The operating point is already supported at 6.92 s by a fresh matched pair captured at 6.90 s. The separate precommitted prediction for 7.10 s becomes assessable only at 12.12 s. That later outcome is not used to justify the earlier operating-point decision.

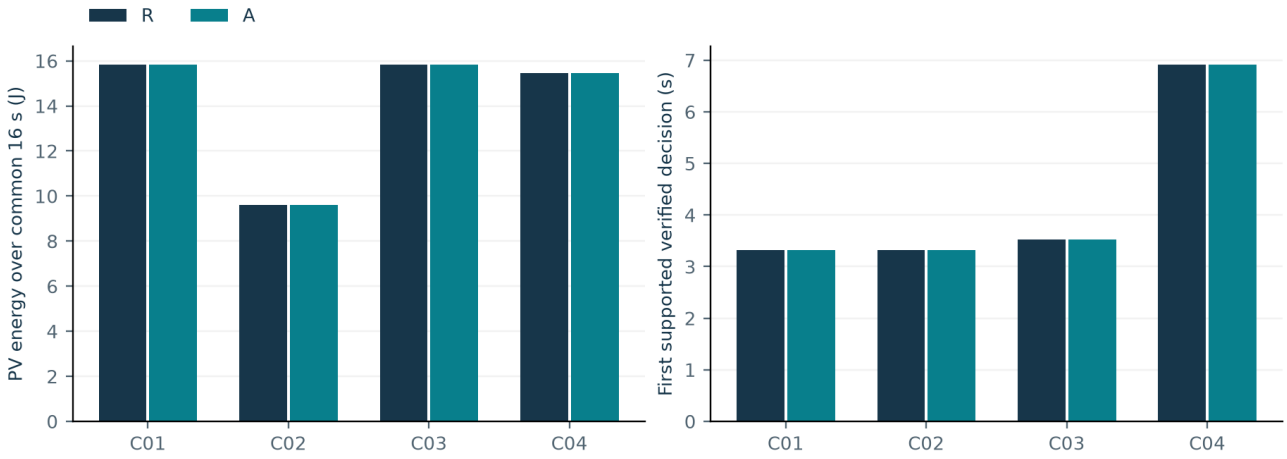
The verification contract

A numerical prediction is committed before dispatch. Its target is an exact advertised capture-grid time. The evaluator looks for that valid voltage sample and a current sample with the same capture time, captured strictly after the relevant successful action terminal. It separately checks the sensor delivery and observation-availability times. Missing, stale or contradictory measurements cannot be replaced by an accepted-command acknowledgement.

Across all eight episodes, **16/16 numerical predictions pass**. These are deterministic voltage-effect checks with a fixed 0.02 V tolerance; they are not calibrated probabilistic predictions of power, fault identity or future weather. Finish-tail samples cannot retroactively justify an earlier success claim.

7. Paired performance and investigation costs

One deterministic episode per case and arm; no confidence intervals or statistical superiority claim.



Case	R / A success	R / A energy (J)	R / A ready (s)	Queries / probe points per arm
C01	True / True	15.84356 / 15.84356	3.32 / 3.32	0 / 0
C02	True / True	9.60474 / 9.60474	3.32 / 3.32	0 / 0
C03	True / True	15.84356 / 15.84356	3.52 / 3.52	1 / 0
C04	True / True	15.46669 / 15.46669	6.92 / 6.92	1 / 3

All episodes reach the same 16 s horizon, including the two-second verification tail. Once ready, both systems preserve the operating point to that horizon. Energy is the simulator's accumulated PV production; it is not net plant service after subtracting real compute, communications or actuator electricity. Frame and camera-motion costs are zero because this batch uses numeric sensing only.

At exactly 16 s, terminal cleanup disconnects the simulated plant after the last integrated interval. Verified operation and accumulated energy refer to the completed episode; the records do not claim continued post-episode generation.

Measured policy-boundary computation totals range from 16.62 to 35.32 ms per episode. These single-run wall-clock measurements include context and output validation and are sensitive to process scheduling. They do not establish a speed ranking. Overruns are rejected on return before dispatch; hostile or non-returning code is not forcibly preempted.

8. What the tests measure, and what they do not

The supplied regression evidence records **875 passed, 0 failed, 0 errors and 0 skipped** in 63.97 s (JUnit suite time). This count is read from the retained XML, not inferred from the number of test functions. The paired batch separately retains eight exact replay checks and the numerical predictions reported above.

Frozen execution runtime: Python 3.11.17; macOS-15.7.4-arm64-arm-64bit; MuJoCo 3.3.7; NumPy 2.4.6; Pydantic 2.14.1. Rendering was disabled for this batch.

Evidence layer	Checks and interpretation
Environment and observation contracts	Existing unit/integration/security regressions check physical-model contracts, sample timing, action execution, data separation and retained record integrity. These are digital correctness checks.
Policy and source checks	Controlled policy tests exercise hold/survey branches, irrelevant/unavailable/conflicting or stale information, bounded clarification, measured target selection and fresh verification. Source text is not allowed to grant permission by substring.
Independent scoring	The scorer reconstructs evidence from original public records, checks action authority and matched settled candidates, and rejects unsupported success. It does not simply trust the policy's final claim.
Paired batch invariants	Same physical config and seed per pair; four 16 s paired horizons; identical local histories before the C03/C04 query; exact transcript replay and metric equality.
Viewer and evidence presentation	Public/evaluator separation, time gating, artifact pairing and source validation are checked by viewer tests. Browser and page-render QA complement data checks.
Not established	No hardware calibration, real sunlight disturbance, noisy repeated performance study, held-out distribution shift, live human/web interaction, continuous learning or open-ended reasoning.

A passing simulation test shows that the specified digital change produces the specified digital effect. Demonstrating agreement with real PV requires synchronized independent physical measurements, uncertainty bounds and model validation against those measurements.

9. The next experiments should test useful differences

Priority	Experiment	Evidence needed before a claim
1	Interrupted investigation and capability changes	Disable a probe, expire service permission, or return conflicting context after planning. Measure supported replanning, false completion, unnecessary actions and safe unresolved outcomes against a strong reference.
2	Noise and difficult sensing	Repeat paired seeds with noise, drift, dropped/late observations and partial actuation. Score detection errors, verification reliability, decision delay and uncertainty calibration where probabilistic outputs exist.
3	Withheld physical combinations	Freeze development and evaluation cases separately. Test source changes combined with transmission or resistance changes without letting the policy read evaluator labels.
4	Real external interaction	Add a bounded human or web session with retained questions, provenance, replies, latency, costs and permission checks. Test whether new information improves a decision, not just whether a response is produced.
5	Real-PV validation	Use an instrumented reference panel/load and independent irradiance and temperature measurements. Apply controlled source/load changes and compare predicted versus measured I-V, power and thermal transients with error bars.
6	Memory and transfer	Only after the single-episode comparison is stable, introduce explicitly versioned learned state and compare later related encounters, forgetting and required human assistance.

Decision at this checkpoint

The first controlled comparison is complete and inspectable. Retain the equal-result finding rather than weakening the reference to manufacture a win. The next scientific step is to preregister a small set of disruptions that require revising a plan, then test whether the explicit goal model improves outcomes or reduces unnecessary intervention.

The report addresses the first five requested development items within this controlled slice: both decision systems, physical/context cases, complete investigation records, paired explanatory visuals and measured evaluation. The broader open-world research programme remains future work.

10. Evidence map and reproducibility

This report is generated from retained records without re-running the simulator. Paths below are relative to the env workspace. The companion JSON manifest contains the full SHA-256 of every consumed input, figure and final PDF.

Frozen experiment

private_eval/paired-investigation-freeze-20261019.json

SHA-256 b536e58b50ee54018ff09b085319d819010ba5cc80846712071694efb4e5e4b9

Paired manifest

runs/paired-investigation-20261019/paired_manifest.json

SHA-256 b6b3e789e4c717dc0aa6587b48c54e428a44d54dab0980d3562f6d4fe2365643

Evaluator summary

private_eval/paired-investigation-20261019/comparison.json

SHA-256 a9e179b21596737af86abff1d61c3ef0d63cd5189e53d212dc0273d552eaa860

Regression JUnit

runs/paired-build-20261019/nonrender.xml

SHA-256 c10c4fe07961eaa6ddb17ba3d06978974c14a35c8cd3318de7621046b3d1a14b

Per-episode records

Each C01-R through C04-A public directory contains investigation.json, policy_audit.json, verification.json, decision_timing.json, public_brief.json and original observation / decision streams. The private companion contains config.json, truth.jsonl, metrics.json and result.json. Hidden schedules, scene geometry and scoring labels remain evaluator-only.

Frozen execution and replay

The experiment records 42 execution-input hashes, runtime versions, both arm mechanisms, exact case configs/corpora, seed, rubric and one repetition per arm. The runner checks that freeze before execution and again before producing the summary. Exact replay reconstructs recorded choices without asking the policy or source to produce new answers. It verifies reproducibility within this declared runtime, not portability to other engines or real hardware.

Read-only report build

scripts/build_paired_report.py consumes --manifest, --freeze and --test-xml, with optional --review; --output and --work must point to a new report and its figure workspace. A changed source or outcome requires a new retained report build, not silently relabelled evidence.